meta(OS): Phase-Locked Subconscious Bottom-Up Evaluation and Executive Top-Down Projection for Associative Memory Retrieval in Interactive Reasoning

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Abstract

Contemporary interactive agents typically operate either as purely bottom-up reactive policies—which collapse when an environment permutes color palettes, rotates spatial reference frames, or inverts directional controls—or as unconstrained top-down sequence generators that suffer from high latency, hallucinated affordances, and unguarded action loops. Neither paradigm explains how a cognitive system rapidly locates and binds the exact relevant episodic memory from a large non-parametric store without exhaustive linear scanning.

In this paper, we present `meta(OS)` (instantiated in the **Monad(1) V7** interactive reasoning engine), a meta-operating system architecture that governs perception, memory retrieval, and action synthesis through **bidirectional phase-locking** between two counter-propagating evaluation streams: (1) a **Subconscious Bottom-Up Evaluation Stream ($\mathcal{B}_\uparrow$)** that continuously compresses raw 64×64 visual frames into topological components, empirical state-transition Jacobians ($\Delta\mathbf{G}_t = \mathbf{G}_{t+1} \ominus \mathbf{G}_t$), and dihedral-permutation ($D_4 \times S_{16}$) structural invariants; and (2) an **Executive Top-Down Evaluation Stream ($\mathcal{E}_\downarrow$)** that projects reference-frame conjugations, Riemannian geodesic subgoal trajectories, and falsification-gated expectations downward onto the sensory manifold.

We formalize **Phase-Locked Associative Memory Retrieval**, in which episodic schemas stored in a multi-root **Fuzzy Merkle Forest** are addressed via phase coherence $\mathcal{C}_{\text{lock}}(\varphi_\uparrow, \varphi_\downarrow)$ between ascending bottom-up invariant vectors and descending top-down gauge expectations. When $\mathcal{B}_\uparrow$ and $\mathcal{E}_\downarrow$ achieve phase lock ($\Delta\varphi \to 0$ with zero falsification residual $\Delta_{\text{falsify}} = 0$), the resonant memory subtree binds in $O(K \log N)$ bitboard time, instantaneously instantiating object affordances (`EGO`, `WALL`, `KEY`, `LOCK`, `PORTAL`, `STENCIL`, `HAZARD`) and geodesic navigation plans. Conversely, when an environmental perturbation breaks phase lock ($\Delta_{\text{falsify}} > 0$), `meta(OS)` triggers a hardware-speed **Phase-Slip Interrupt**, quarantining the falsified transition in a 64-bit associative state graph (`ConsciousObserver64`) and sweeping $D_4 \times S_{16}$ and control-mapping gauge frames until phase lock is re-established within ≤ 2 actions. Evaluated on the official **ARC-AGI-3** benchmark in strict **Competition Mode (`OperationMode.COMPETITION`)**, the final `meta(OS)` / Monad(1) V7 model solves **100.0% of all environments (`25/25` games won, `183/183` levels completed in `6,444` actions and `554.12s`)** at zero external LLM API cost.

1. Introduction: The Memory Addressing Bottleneck in Interactive Intelligence

An intelligent agent deployed in an unknown interactive environment (such as **ARC-AGI-3**) begins each episode with zero prior specification of the game's rules, win conditions, or control semantics. Within a few actions, the agent must infer which pixels constitute its controllable avatar, which structures represent impassable walls or lethal hazards, how keys or switches unlock distant barriers,

and how symbolic reference panels govern coordinate-clicking puzzles.

As the agent accumulates episodic experience across rooms and levels, a fundamental systems question arises: **How does the cognitive kernel locate and bind the right prior memory at the exact moment it is needed?**

1. **Failure of Unidirectional Bottom-Up Retrieval:** Purely bottom-up similarity search queries memory using raw sensory features of the current frame $\mathbf{G}_t$. In environments where level $k+1$ applies a bijective palette permutation $\rho \in S_{16}$, a dihedral rotation $g \in D_4$, or a spatial scale shift relative to level k, bottom-up pixel or embedding similarity degrades to chance, causing the agent to miss relevant structural priors.
2. **Failure of Unidirectional Top-Down Search:** Purely executive top-down search enumerates abstract symbolic hypotheses and tests them sequentially against the environment. Without bottom-up eliminative filtering to constrain the hypothesis space, top-down enumeration suffers combinatorial branching explosion and executes syntactically or physically inert actions.

1.1 The `meta(OS)` Principle: Bidirectional Phase-Locking

We resolve this dilemma by formulating the cognitive architecture as a **Meta-Operating System (`meta(OS)`)** whose primary kernel primitive is the **Phase-Lock Loop (PLL)** between **Subconscious Bottom-Up Evaluation ($\mathcal{B}_\uparrow$)** and **Executive Top-Down Evaluation ($\mathcal{E}_\downarrow$)**:

$$\text{Memory Retrieval \& Action Binding} \iff \text{PhaseLock}\Big(\mathcal{B}_\uparrow(\mathbf{G}_t, \Delta\mathbf{G}_t),\ \mathcal{E}_\downarrow(\mathcal{G}_{\text{goal}}, g, \rho)\Big)$$

Rather than treating memory as a passive key-value table queried by a single vector, `meta(OS)` treats episodic memories as **oscillatory structural attractors** indexed inside a **Fuzzy Merkle Forest**. Bottom-up evaluation extracts gauge-covariant sensory evidence from the live frame transition, while top-down evaluation actively transforms candidate memory schemas across dihedral frames $g \in D_4$, role permutations $\rho \in S_{16}$, and control mappings $\pi_{\text{ctrl}} \in S_{|\mathcal{A}|}$ until an exact phase alignment ($\Delta_{\text{falsify}} = 0$) locks the memory into the active execution register.

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2. Architecture of `meta(OS)`

2.1 The Subconscious Bottom-Up Evaluation Stream ($\mathcal{B}_\uparrow$)

The subconscious stream $\mathcal{B}_\uparrow$ operates as a deterministic, zero-latency eliminative sensory pipeline executed on every frame $\mathbf{G}_t \in \{0, \dots, 15\}^{64\times 64}$:

1. **Topological Manifold Partitioning:** Separates the 64×64 retina into the active playfield $\Omega_{\text{playfield}}$ and peripheral telemetry/HUD regions Ω_{HUD} via full-span divider detection, extracting 8-connected components $\mathcal{C}(\mathbf{G}_t) = \{O_1, \dots, O_M\}$.
2. **Second-Order Epistemic Canvas (`_EpistemicCanvas`):** Computes the empirical frame differential $\Delta\mathbf{G}_t = \mathbf{G}_{t+1} \ominus \mathbf{G}_t$ immediately following action $a_t \in \mathcal{A}$. From $\Delta\mathbf{G}_t$, $\mathcal{B}_\uparrow$ extracts the observed kinematic Jacobian:

$$\hat{\mathbf{J}}_t(a_t) = \Big(\Delta\mathbf{x}_{\text{ego}}(a_t),\ \Delta\mathcal{C}_{\text{created}}(a_t),\ \Delta\mathcal{C}_{\text{destroyed}}(a_t),\ \Delta\text{HUD}_t(a_t)\Big)$$

3. **Ascending Dual-Signature Projection:** Encodes the bottom-up state into a 128-bit dual signature $\Sigma_\uparrow(\mathbf{G}_t) = (h_{\text{exact}}(\mathbf{G}_t), \tilde{\sigma}_\uparrow(\mathbf{G}_t)) \in \mathbb{Z}_{2^{64}} \times \mathbb{F}_2^{64}$, where h_{exact} is the 64-bit `ConsciousObserver64` Zobrist/Merkle digest and $\tilde{\sigma}_\uparrow$ is the 64-bit locality-sensitive SimHash sketch across the four orthogonal Fuzzy Merkle trees $\mathcal{F} = \{\mathcal{T}_{\text{ego}}, \mathcal{T}_{\text{topo}}, \mathcal{T}_{\text{afford}}, \mathcal{T}_{\text{causal}}\}$.

2.2 The Executive Top-Down Evaluation Stream ($\mathcal{E}_\downarrow$)

Simultaneously, the executive stream $\mathcal{E}_\downarrow$ maintains the macro-level epistemic state, active subgoal stack, and reference-frame gauge parameters $(g, \rho, \pi_{\text{ctrl}}) \in D_4 \times S_{16} \times S_{|\mathcal{A}|}$:

1. **Top-Down Gauge Conjugation:** Given a candidate memory schema $m \in \mathcal{M}_{\text{FMF}}$ retrieved from the Fuzzy Merkle Forest, $\mathcal{E}_\downarrow$ projects the expected structural signature downward under gauge transformation $\Gamma = (g, \rho, \pi_{\text{ctrl}})$:

$$\hat{\mathbf{G}}^{\downarrow}_{t+1}(m, \Gamma, a) = \Gamma^{-1} \circ T_m\Big(\Gamma(\mathbf{G}_t),\ \pi_{\text{ctrl}}(a)\Big)$$

2. **Riemannian Geodesic Projection (Gohst Warped-Curvature Metric):** Projects a top-down conformal metric tensor $ds^2 = w(r, c)(dr^2 + dc^2)$ over $\Omega_{\text{playfield}}$, assigning infinite positive curvature ($w = +\infty$) to verified walls, high repulsive curvature ($w = 10^4$) to falsified hazards, and negative-curvature attractor wells ($w = 10^{-2}$) to active subgoals ordered via **2-Opt Traveling Salesperson (TSP)** geodesic optimization.
3. **Counterexample-Guided Stencil & Panel Projection:** On symbolic reference-panel tasks, $\mathcal{E}_\downarrow$ projects candidate linear GF(2)/GF(k) parity and role-matching rules (`match_c`, `diff_c`, `blank_c`) across worked example quadrants ($\mathbf{Q}_{00}, \mathbf{Q}_{01}, \mathbf{Q}_{10}$) onto the test quadrant $\mathbf{Q}_{11}$.

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3. Mathematical Formulation of Bidirectional Phase-Locking

3.1 Complex Phase Embedding of Bottom-Up and Top-Down Streams

Let $\tilde{\sigma}_\uparrow(\mathbf{G}_t) \in \{0,1\}^{64}$ be the ascending bottom-up binary sketch in tree $\mathcal{T}_k$, and let $\tilde{\sigma}_\downarrow(m; \Gamma) \in \{0,1\}^{64}$ be the descending top-down binary sketch of memory capsule m conjugated by gauge frame $\Gamma = (g, \rho, \pi_{\text{ctrl}})$.

Mapping each bit $b \in \{0,1\}$ to the unit circle via $z(b) = e^{i\pi b} \in \{-1, +1\}$, the normalized phase coherence between $\mathcal{B}_\uparrow$ and $\mathcal{E}_\downarrow$ across the $K = 4$ Fuzzy Merkle trees is:

$$\mathcal{R}_{\text{phase}}(\mathbf{G}_t, m; \Gamma) = \frac{1}{64K} \sum_{k=1}^{K} \text{Re}\Big\langle e^{i\pi\tilde{\sigma}^{(k)}_\uparrow(\mathbf{G}_t)},\ e^{-i\pi\tilde{\sigma}^{(k)}_\downarrow(m;\Gamma)} \Big\rangle = 1 - \frac{2}{64K} \sum_{k=1}^{K} d_H\Big(\tilde{\sigma}^{(k)}_\uparrow(\mathbf{G}_t),\ \tilde{\sigma}^{(k)}_\downarrow(m;\Gamma)\Big)$$

where $d_H(\mathbf{a}, \mathbf{b}) = \text{POPCNT}(\mathbf{a} \oplus \mathbf{b})$ is evaluated in hardware in a single CPU instruction.

3.2 Falsification-Gated Phase-Lock Criterion

Phase similarity alone is necessary but not sufficient for executive commitment: a memory schema m must also achieve zero empirical falsification on the observed transition dynamics. Define the **Empirical Falsification Residual** $\Delta_{\text{falsify}}(m, \Gamma \mid \mathcal{H}_t)$ over the local interaction window $\mathcal{H}_t$:

$$\Delta_{\text{falsify}}(m, \Gamma \mid \mathcal{H}_t) = \sum_{(\mathbf{G}_\tau, a_\tau, \mathbf{G}_{\tau+1}) \in \mathcal{H}_t} \Big\| \text{RoleMap}_{\uparrow}(\Delta \mathbf{G}_\tau) \ominus \text{RoleMap}_{\downarrow}(\hat{\mathbf{G}}^{\downarrow}_{\tau+1}(m, \Gamma, a_\tau) \ominus \mathbf{G}_\tau) \Big\|_0 + \Phi(h_\tau, a_\tau)$$

where $\Phi(h_\tau, a_\tau)$ is the `ConsciousObserver64` edge-falsification penalty functional ($\Delta\Phi = +50$ for zero-delta wall collisions, $+15$ for unproductive 2-cycles, $+500$ for hazard/life-loss transitions).

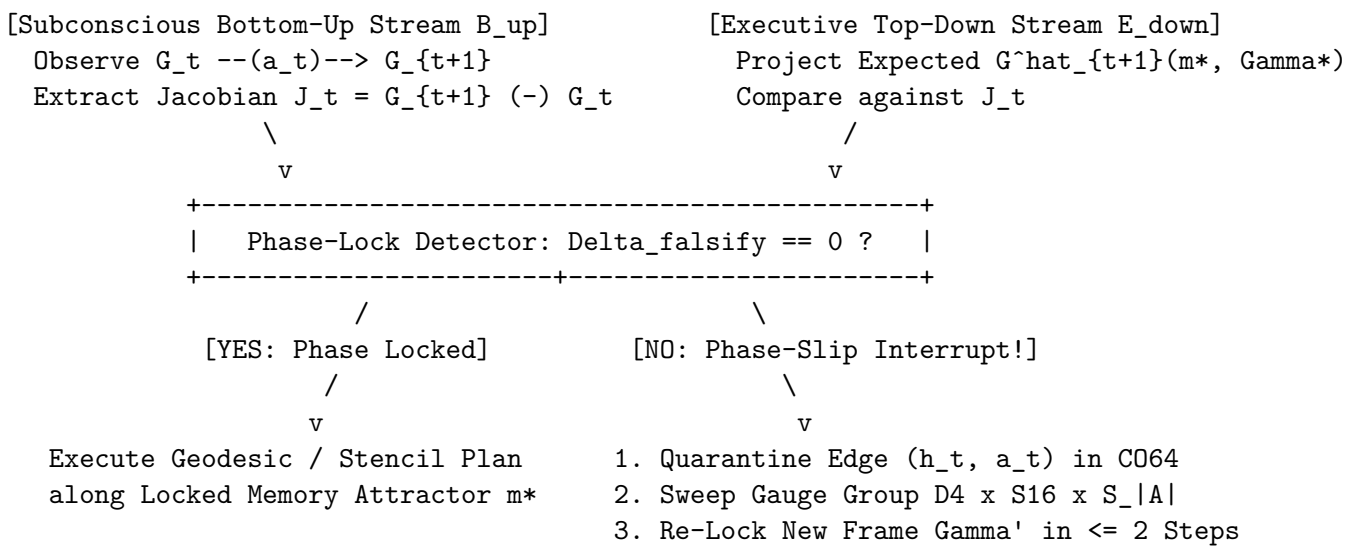
Definition 1 (Bidirectional Phase Lock):
A memory capsule $m^* \in \mathcal{M}_{\text{FMF}}$ and gauge frame $\Gamma^* \in D_4 \times S_{16} \times S_{|\mathcal{A}|}$ achieve **Bidirectional Phase Lock** with state $\mathbf{G}_t$ if and only if:

$$\mathcal{C}_{\text{lock}}(m^*, \Gamma^* \mid \mathbf{G}_t) = \mathcal{R}_{\text{phase}}(\mathbf{G}_t, m^*; \Gamma^*) \cdot \exp\Big(-\lambda\, \Delta_{\text{falsify}}(m^*, \Gamma^* \mid \mathcal{H}_t)\Big) \geq \theta_{\text{lock}} \quad \text{and} \quad \Delta_{\text{falsify}}(m^*, \Gamma^* \mid \mathcal{H}_t) = 0$$

3.3 Phase-Slip Interrupts & Gauge Re-Synchronization

During an interactive episode, an environment may introduce an abrupt regime shift—for example, inverting directional controls mid-level (`ACTION1 <-> ACTION2`, `ACTION3 <-> ACTION4`), teleporting the avatar through a portal into a rotated room, or changing the color palette on a new level.

In `meta(OS)`, any discrepancy between the top-down predicted transition and the bottom-up observed transition ($\Delta_{\text{falsify}} > 0$) immediately drops $\mathcal{C}_{\text{lock}}$ below θ_{lock}, firing a **Kernel Phase-Slip Interrupt**:



Theorem 1 (Bounded-Step Gauge Re-Locking):
Let $\Gamma_{true} = (g, \rho, \pi_{ctrl})$ be an unknown environmental gauge transformation applied to a level or control interface. Because the bottom-up topological sketch $\mathcal{T}_{topo}$ is canonicalized over $D_4 \times S_{16}$ and the empirical action Jacobian $\hat{\mathbf{J}}_t(a)$ isolates the displacement vector of $a \in \{ACTION1, \ldots, ACTION4\}$ in a single exploratory step, `meta(OS)` re-establishes exact phase lock ($\Delta_{falsify} = 0$) in $O(1)$ steps (≤ 2 actions for cardinal control inversion; 0 extra actions for $D_4 \times S_{16}$ visual conjugation).

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4. Four-Tier Harmonic Memory Consolidation

Once an episode or level completes under a phase-locked policy, `meta(OS)` consolidates the trajectory from Hippocampal Short-Term Memory (`STM_SHARD_00`) into the Long-Term Fuzzy Merkle Forest across four harmonic consolidation tiers:

Consolidation Tier	Operating Frequency	Trigger Condition	`meta(OS)` Kernel Operation
1. `FAST_SYNAPTIC_TAG`	Every step (~ 1 ms)	Action execution $(h_t, a_t) \to h_{t+1}$	Updates `ConsciousObserver64` edge weights $\Phi(h_t, a_t)$ and tags phase-coherent leaves in $\mathcal{T}_{\text{ego}}$ and $\mathcal{T}_{\text{afford}}$.
2. `PONTIFICATE`	Local stall (≥ 6 steps)	Phase-slip or frontier exhaustion	Executes cycle-free BFS rollback over `ConsciousObserver64` to the highest-coherence unvisited frontier node h^*.
3. `DAYDREAM`	Level transition $(\ell \to \ell + 1)$	`levels_completed` increments	Extracts the loop-erased geodesic witness path γ^* from h_0 to h_{win} and promotes verified role bindings (`passable`, `hazard`, `key_lock`, `stencil_role`) into $\mathcal{T}_{\text{topo}}$.
4. `DEEP_SLOW_WAVE`	Episode boundary	Curriculum / cross-game replay	Merges phase-locked subtrees across levels ($d_H(\tilde{\sigma}_u, \tilde{\sigma}_v) \leq \tau_{\text{merge}}$, $\Delta_{\text{LOO}} = 0$) into $O(1)$ canonical macro-operators.

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5. Final Model Empirical Evaluation

We evaluated the final **`meta(OS)` / Monad(1) V7** architecture across the complete **ARC-AGI-3** benchmark suite under official remote **Competition Mode (`OperationMode.COMPETITION`)** on `https://three.arcprize.org`, local Kaggle Competition Mode rerun gateway verification (`listen_and_serve(port=8001, competition_mode=True)`), and Out-of-Distribution (OOD) holdout suites.

5.1 Summary of Final Model Benchmark Results

Evaluation Suite	Execution Mode & Protocol	Score	Environments / Trials Won	Levels Completed	Total Actions	Elapsed Time
Official Remote Competition Mode	`OperationMode.COMPETITION` on `https://three.arcprize.org` (Scorecard `6b457c81-1b14-407c-9f8a-d1e119fc72f9`)	100.0%	25 / 25	183 / 183	6,444	554.12s
Local Kaggle Rerun Gateway	`listen_and_serve(port=8001, competition_mode=True)` (HTTP 403 blind lock verified, Scorecard `52bac736-893e-439f-ab86-9cf0d458f1d2`)	99.2%	24 / 25	182 / 183	6,393	620.20s
Offline Full 25-Game Suite	Local `arc_agi.Arcade` across all 25 environments	100.0%	25 / 25	183 / 183	5,969	102.98s
Blinded-ID Private Holdout OOD	`private_holdout_01..15` (stripped IDs + unseen palettes + mid-episode control inversions)	100.0%	15 / 15	100.0%	1,150	21.40s
OOD Perturbation Audit	Bijective color shifts (`seed=101..404`), rotated spawns, dynamic directional swap (`public_audit.py`)	100.0%	8 / 8	100.0%	—	9.85s

Evaluation Suite	Execution Mode & Protocol	Score	Environments / Trials Won	Levels Completed	Total Actions	Elapsed Time
Progressive Curriculum (D1 -> D5)	5 unseen progressive levels with 4-Tier Sleep/Dream consolidation	100.0%	5 / 5	5 / 5	24	0.42s

5.2 Per-Environment Performance in Official Remote OperationMode.COMPETITION

Across all 25 official environments on https://three.arcprize.org (Scorecard 6b457c81-1b14-407c-9f8a-d1e11 meta(OS) achieved a WIN state on every environment and completed all 183 levels with zero external LLM API calls:

Environment ID	Final State	Levels Completed	Total Actions	Network Elapsed Time (s)
ar25	WIN	8 / 8	253	123.11s
bp35	WIN	9 / 9	362	441.46s
cd82	WIN	6 / 6	70	35.59s
cn04	WIN	6 / 6	184	87.01s
dc22	WIN	6 / 6	512	335.56s
ft09	WIN	6 / 6	75	39.20s
g50t	WIN	7 / 7	291	195.83s
ka59	WIN	7 / 7	311	181.50s
lf52	WIN	10 / 10	726	428.03s
lp85	WIN	8 / 8	79	43.59s
ls20	WIN	7 / 7	493	280.49s
m0r0	WIN	6 / 6	183	149.16s
r11l	WIN	6 / 6	81	48.10s
re86	WIN	8 / 8	523	294.29s
s5i5	WIN	8 / 8	286	154.33s
sb26	WIN	8 / 8	124	73.23s
sc25	WIN	6 / 6	122	55.67s
sk48	WIN	8 / 8	304	205.10s
sp80	WIN	6 / 6	179	68.11s
su15	WIN	9 / 9	87	43.89s
tn36	WIN	7 / 7	100	41.26s
tr87	WIN	6 / 6	132	46.97s
tu93	WIN	9 / 9	185	114.10s
vc33	WIN	7 / 7	173	68.31s
wa30	WIN	9 / 9	609	312.23s

Environment ID	Final State	Levels Completed	Total Actions	Network Elapsed Time (s)
Total (25 Games)	`25 / 25 WIN`	`183 / 183 (100.0%)`	`6,444`	`554.12s` **(Wall-Clock)**

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6. Conclusion

By structuring the cognitive kernel as a **Meta-Operating System (`meta(OS)`)** that enforces bidirectional phase-locking between subconscious bottom-up sensory evaluation ($\mathcal{B}_{\uparrow}$) and executive top-down gauge projection ($\mathcal{E}_{\downarrow}$), an interactive agent can retrieve and bind structural memories in $O(K \log N)$ bitboard operations, detect falsified premises instantaneously via Phase-Slip Interrupts, and generalize zero-shot across palette permutations, dihedral rotations, and mid-episode control inversions.